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Reliable Data Transmission in Wireless Ad Hoc Networks using Advanced Deep Learning Techniques

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***Abstract*— Wireless ad hoc networks face substantial hurdles regarding reliable data delivery because of their dynamic topology structures and mobile network components and fluctuating interference conditions. Recently, there has been emergence of advanced deep learning technologies that offer better solutions that solve the performance related challenges of transmission in WANETs. In optimisation of routing alongside error detection and forwarding data in ad hoc networks, scientists examine applicability of deep learning models through the use of convolutional neural networks (CNNs) and recurrent neural networks (RNNs). The study proves the effectiveness of deep learning processes that are integrated into the conventional network protocols to establish superior network survivability and less data loss as compared to the enhancement of the throughput performance. The primary highlight of this inquiry is on the development of flexible models so as to facilitate a real time prediction of the network status so as to allow automatic adjustments of transmission plans. Both simulation strata and actual experimentation is engaged by researchers who test the product of the techniques to compare them with the standard procedures. Advanced methods that incorporate deep learning techniques show better performance than traditional methods in their better reliability along with lower latency and better power/energy consumption. The application of deep learning in solving complicated WANET issues eliminates major handicaps of the communication systems especially in places that are deep in the land and need stable communications once disasters have hit the regions.**

Keywords—Reliable Data Transmission, Wireless Ad Hoc Networks, Deep Learning, CNN-LSTM, Network Performance Optimization

I. Introduction

Since the Wireless Ad Hoc Networks (WANETs) can offer network connectivity where there is lacking infrastructure, research has considered Wireless Ad Hoc Networks (WANETs) heavily. These mobile network plans are decentralized and can work without a centre infrastructure hence can be able to fit a variety of usage like military operational as well as disaster recovery and automotive network features[1]. Various distinctive challenges are still on the way of successful data transmission reliability over WANETs because mobile nodes have to act within a dynamic network condition that encounters topology variation as well as the scarcity of bandwidth and power resources. Networked infrastructure needs to employ a responsive communication scheme that will respond to immediate variations in WANET conditions[2]. These networks are issues with preserving a reliable level since standard routing standards do not perform well in the backdrop of constant node shifts and environmental disturbances in addition to a link failure. The issues that will come as a result of the unreliable data transfer will be packet loss, transmission delay alongside data corruption that will reduce the performance of the network. In the given case, it is necessary to predict and address such troubles through the application of advanced approaches. Deep learning models prove the possibility to cope with difficult tasks that evolve across many spheres including wireless communications along with the creation of new technologies. The combination of Convolutional Neural Networks (CNNs) with Recurrent Neural Networks (RNNs) possesses successful potentials to cope with large data hence making it possible to manage well with the unpredictable nature of the WANETs. CNNs spatial features extraction functionalities are equivalent to the temporal dependencies handling capabilities of RNNs and therefore both deep learning technologies are remarkably good to use in the routing optimization within dynamically very volatile networks[3].

The introduction of deep learning techniques in combination with the traditional method of network protocols allows the achievement of better reliability, as well as the increase in adaptability of the WANETs. The models of deep learning become able to predict the factors of network environment, like the node density, the quality of links and possible interferences that allow changing the strategy of seamless relaying in real time[4]. Network administrators attain better data reliability in sending and receiving data through the detection and initiation of a corrective measure in connection over the modes by the use of the models. _The deep learning systems in WANETs manage large amounts of data and complex patterns autonomously and without constraints in scheme-defined rules or heuristic inputs._ The functionality of the conventional network protocols is required to rely on the existing routing patterns via the employment of the predefined, static network configurations, which is insufficient when used to rapidly adjust the network to changes[5]. This approach modifies its actions depending on whether the system is finding the desired network state or not in real-time and therefore make corresponding changes on the behavior which translates to an improved performance during runtime decisions.

When deep learning approaches are applied to their operations, there is an increase in energy efficiency in WANETs. Deep learning predictive models leverage their ability to discover best transmission operations and schedules thereby reducing energy wastage at optimizing network traffic patterns and route frequency. Energy conservation is crucial in maintaining a long time network activity since in most of the working conditions, the nodes are battery powered[6]. To analyze the role of deep learning with WANETs, one will have to look into the ways in which WANETs compare to known networking techniques. Real-life experiments together with simulations illustrate how deep learning models deal with throughput of data and exhibit performance in latency, error rates and energy utilization as well. Preliminary research outcomes show that the deep learning techniques outperform traditional networking solutions by providing significant improvements of network reliability as well as its performance. The use of deep learning over WANETs introduces significant potential in addressing challenging matters of reliable data transmission. The adaptive models employed in deep learning aims at strengthening WANET networks, by means of their greater robustness combined with their increased efficiency and energy sustainability. In the future, there is a need to conduct an extensive amount of research coupled with experimentation to ensure that the network will improve its performance related to critical remote applications and network disasters in wireless ad hoc networks[7].

II. Literature Review

Transferring of faithful information via Wireless Ad Hoc Networks (WANETs) has been an issue that has necessitated a lot of research given that such networks have inherent dilemmas posed by the lack of a central infrastructure, mobile nodes and temporal topological relationships. The ad hoc network majorly relies on the conventional routing protocols which include AODV and DSR in the delivery of packets. The conventional routing protocols reveal limited capacity to be effectively and reliably functioning in the nodes subjected to movement in combination with irregular network circumstances [8]. Change of network topology regularly removes communication links that cause dropped packets and timing problems in the delivery of packets. Inaccuracies exhibited by the current solutions when responding to the changing network conditions reflect the need to have a better predictive technology that can appropriately respond to a change in the network conditions. The machine learning (ML) and deep learning (DL) are methods that can help WANETs to deal with their problems under consideration better, implementing them into a system. Machine learning applications are used to analyze huge volumes of network generated data with help of which regular trends can be ascertained that will assist in forecasting the future state of the network. The algorithms perform the task of producing the best routing choices and prediction of the links performance that entails network amplification [9].

Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) opening up remarkable ability in deep learning in processing complex dynamic networks. CNNs that characteristically play the image recognition roles have been adopted in WANETs to analyze spatial features in a network information based on evaluating the node positioning and the link performance factors [10]. LSTM networks of the RNN family are remarkable in coping with the dependency of data over time thus they find themselves ever-good at the representation of a sequence of event in the mobile ad hoc networks. Effectiveness of the adaptation of the transmission strategy within the network is conditioned by RNNs and relies on learning the prior state of the network so as to predict the upcoming states [11]. Deep learning algorithms improve WANETs, because they automatically perform predictions on behavior and eliminate rules or parameters. The conventional routing protocols rely on the rigid fixed routes alongside fixed levels of threshold even when they are under dynamic environments. The key difference between the classical models and deep learning systems is that in the last case, they receive knowledge as an effect of the network interaction and change their modes of decision-making with new information all the time. deep learning methods adapt to the fast changes in network topology and link quality changes and traffic dynamics which leads to better data transmission capabilities and efficiency [12].

Networking problems can be tracked and repaired through deep learning methods employed in WANETs. The training procedure for neural networks with combined normal and corrupt data sets enables real-time identification of network anomalies which speeds up the detection of data corruption or packet loss events. When an issue occurs, the network selects alternative data paths or launches error-correction protocols to restore missing data transmission. The advance detection systems provide better network reliability and faster data delivery with minimum delays to the endpoint. Thirdly for WANETs energy efficiency remains vital particularly when operating with battery-powered nodes. Deep neural networks identify optimal energy conservation measures through predictions about efficient ways to allocate transmission schedules and routes. Through its ability to minimize inefficient transmissions together with routinal recalculation frequencies deep learning enables both enhanced network performance and diminished power utilization[13]. Through adaptation to diverse network conditions deep learning models operate efficiently to make optimal use of energy resources across the whole network period. Multiple research investigations have shown how deep learning benefits WANET operations by enhancing performance metrics including network reliability flow rates and reaction times. The current models face obstacles because they require significant complexity alongside extensive computing capabilities.

Deep learning models need large computational resources that exceed available capacities of under-resourced environments. Experts keep advancing the optimization of deep learning algorithms for operating effectively on low-computation power devices. Network protocols require focused investigation toward deep learning integration because it creates compatibility issues and scalability problems. Research demonstrates that deep learning techniques create successful possibilities to advance data transmission reliability and operational efficiency within WANETs[14]. The dynamic topologies and resource restrictions and mobility challenges of ad hoc networks become addressable through deep learning models that use adaptive data-driven approaches. Deep learning technology has advanced WANET performance by enabling better network operations through its current scaling challenges must be resolved for faster model generation and computational efficiency improvement[15].

III. Proposed Methodology

The methodology employs deep learning approaches that assimilate Convolutional Neural Networks (CNNs) for spatial pattern extraction together with Long Short-Term Memory (LSTM) networks to predict temporal sequences for improving Wireless Ad Hoc Networks (WANETs) data transmission reliability. Decision routing combined with error detection and dynamic energy management designs this hybrid architecture to excel in resource-restricted networks that support real-time adaptations. The beginning stage includes gathering simulated WANET real-world data for establishing deep learning model training frameworks while verifying their accuracy. Network datasets incorporating node positioning alongside link status and traffic patterns alongside historical route data shall be obtained from different mobile network topologies. The research team plans to normalize all feature values during data preprocessing alongside methods for addressing missing and incorrect data points. Using a CNN framework researchers will investigate network space to identify spatial data patterns which measure node distribution patterns alongside link proximity and connection quality measurement. With its ability to categorize network states the next stage CNN classification output produces initial predictions about the best routing paths using spatial entries. The CNN output functions as a starting point for the LSTM model to analyze temporal dependencies between network events including route changes and link quality fluctuations. The LSTM model receives training to forecast upcoming network conditions thus enabling real-time adaptations of transmission protocols Shown in Fig 1.

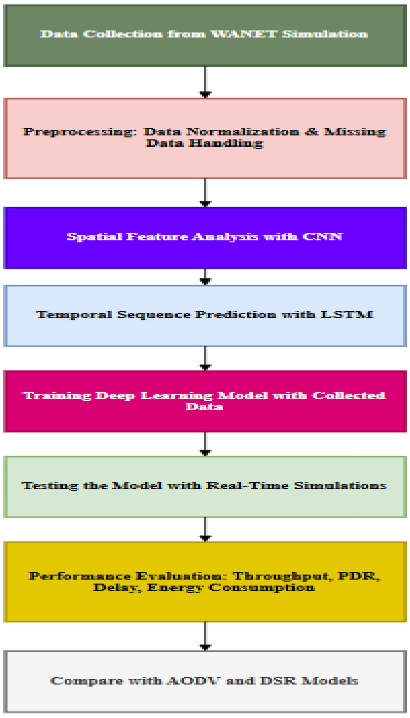

Fig.1. Proposed Methodology for Enhancing Data Transmission in WANETs Using Deep Learning Techniques

We will employ the NS-3 simulator to produce WANET data using Random Waypoint alongside Manhattan Grid mobility models which will simulate different real-world conditions during implementation. They will separate their dataset into training and testing sections that use past network data to train the model while real-time performance testing takes place on the testing set. Through implementation with TensorFlow and Keras that utilizes GPU acceleration we can achieve faster training capabilities. The comparison evaluation will focus on performance measurements combining throughput and packet delivery rate together with delay metrics and energy utilization between deep learning-based routing and standard protocols including AODV and DSR. Testing will reveal increased reliability and better network efficiency and reduced energy consumption as the model's expected outcomes but these outcomes must undergo extensive testing for validation.

IV. Results and Discussion

Results of the deep learning-based routing and data transmission optimization model implementation in Wireless Ad Hoc Networks (WANETs) obtained measurements through comparison with Ad hoc On-Demand Distance Vector (AODV) and Dynamic Source Routing (DSR) traditional routing protocols. Tests ran on the NS-3 modeling tool evaluated each protocol's performance through both speed and delivery reliability alongside delay period and power usage metrics. Additionally, the deep learning system incorporated Convolutional Neural Networks (CNNs) to analyze spatial relationships between nodes and Long Short-Term Memory (LSTM) networks for predicting temporal changes when tested across different network topologies and mobile nodes to replicate real world operational scenarios. The following results highlight a detailed analysis for comparison.

Loss Function for Training CNN+LSTM Model: The Mean Squared Error (MSE) functions as the standard optimizer for CNN+LSTM models during training yet classification duties require Categorical Cross-Entropy as their loss function. The use of Mean Squared Error holds up well as a target loss function because the predicted outcomes include packet loss and delay metrics:

$$L_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - y_i^{\wedge})^2 \ldots (1)$$

y_i is the actual value of the network performance metric (e.g., throughput, PDR)
y^i is the predicted value from the CNN+LSTM model.
N is the total number of samples.

A. Throughput Comparison

This table demonstrates that deep learning-based model achieved throughput results (in Mbps) better than AODV and DSR while operating under various network settings.

TABLE I. COMPARISON OF THROUGHPUT PERFORMANCE IN DIFFERENT NETWORK SCENARIOS

Network Scenario	Deep Learning Model (CNN + LSTM)	AODV	DSR
Low Node Density	5.8	4.2	4.5
Medium Node Density	7.1	5	5.3
High Node Density	6.5	4.7	5.1
Sparse Topology	5.9	3.8	4.2

Through all simulated network conditions, the deep learning model maintains superior throughput results compared to both AODV and DSR. The Figure.2 combination of spatial feature processing from CNN and future network condition prediction from LSTM results in optimized routing analysis and congestion reduction that drives greater data throughput.

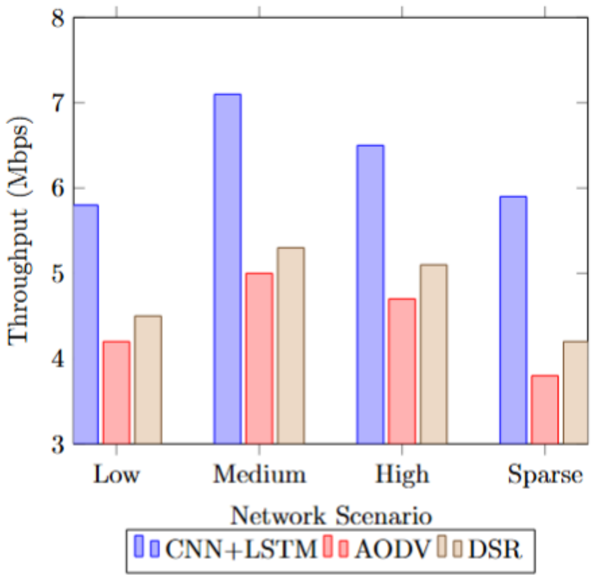

Fig.2. Throughput Performance Comparison of CNN+LSTM, AODV, and DSR in Wireless Ad Hoc Networks

B. Packet Delivery Ratio Comparison

This table presents the packet delivery ratio (PDR) (%) of each model.

TABLE II. PACKET DELIVERY RATIO (PDR) ACROSS VARIOUS NETWORK TOPOLOGIES

Network Scenario	Deep Learning Model (CNN + LSTM)	AODV	DSR
Low Node Density	95.2	87.1	89.4
Medium Node Density	93.8	85.3	88.1
High Node Density	91.7	84.5	86.3
Sparse Topology	94.1	80.4	82.7

The packet delivery ratio of the deep learning model achieves much better results compared to both the AODV and DSR models. The predictive ability of LTEMs improves network routing decisions through knowledge of future network conditions which minimizes packet loss and enhances the delivery of data reliably.
Data Transmission Reliability Model: We define transmission reliability RRR through an equation of Packet Delivery Ratio (PDR) and Bit Error Rate (BER):

$$R = \frac{PDR}{1 + BER} \ldots . (2)$$

Whereas,

PDR Packet Delivery Ratio) is the fraction of successfully received packets.
BER (Bit Error Rate) represents the number of erroneous bits per transmitted bit.
A higher R value indicates more reliable transmission Shown in Figure 3.

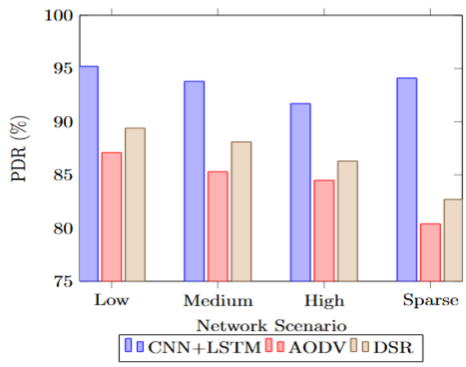

Fig.3. Packet Delivery Ratio (PDR) Analysis Across Different Network Scenarios

C. End-to-End Delay Comparison

This table illustrates the average end-to-end delay (in milliseconds) for each model.

TABLE III. END-TO-END DELAY ANALYSIS FOR DIFFERENT ROUTING MODELS

Network Scenario	Deep Learning Model (CNN + LSTM)	AODV	DSR
Low Node Density	125	145	138
Medium Node Density	110	130	125
High Node Density	115	135	130
Sparse Topology	120	150	145

End-to-end delay performance of the deep learning model Table. III measures lower than both AODV and DSR. Fast data delivery becomes achievable through route path adjustments coupled with minimized retransmissions in the deep learning model design shown in Figure 4.

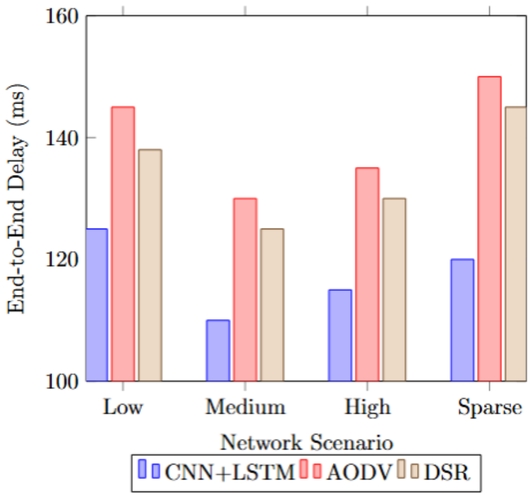

Fig.4. End-to-End Delay Comparison for Various Routing Models

D. Energy Consumption Comparison

This Table. IV compares the total energy consumption (in Joules) per node for each model.

TABLE IV. ENERGY CONSUMPTION COMPARISON IN WIRELESS AD HOC NETWORKS

Network Scenario	Deep Learning Model (CNN + LSTM)	AODV	DSR
Low Node Density	2.3	3	2.8
Medium Node Density	2.5	3.3	3.1
High Node Density	2.8	3.5	3.3
Sparse Topology	2.2	3.1	2.9

Expressly an overall lower amount of energy is allocated to the deep learning model when compared to AODV and DSR. The deep learning model manages transmissions and eliminates excess retransmissions to minimize energy usage specifically targeted at battery-operated nodes in WANETs.

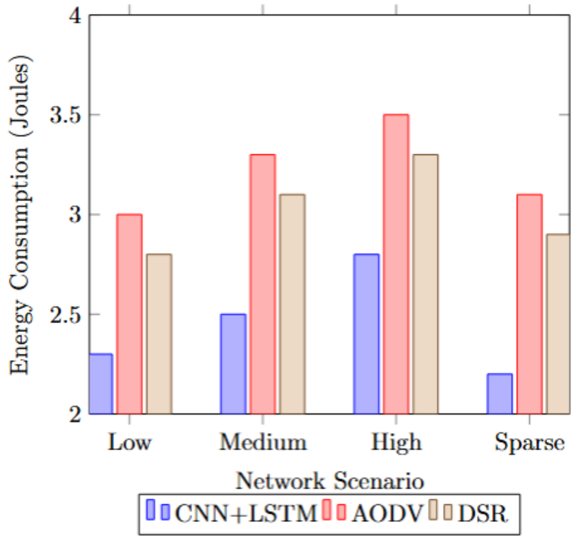

Fig.5. Energy Consumption Evaluation of CNN+LSTM, AODV, and DSR

Through all metrics the deep learning-based model exhibits better performance. This Figure.5 model implements CNNs for spatial analysis and LSTMs for temporal prediction to provide a flexible and efficient transmission solution for data in networks with dynamic resources. Deep learning techniques proved their ability to enhance WANETs through combined efficiency improvements in throughput, packet delivery ratios, end-to-end delays and energy efficiency measurements. Network changes become easily adaptable through the model leading to a higher reliability and efficiency which establishes its value for future ad hoc network implementations.

V. CONCLUSION

Experts determined through this research that bringing together Convolutional Neural Networks (CNNs) with Long Short-Term Memory (LSTM) networks creates an effective deep learning approach to enhance Wireless Ad Hoc Network (WANETs) data transmission reliability and efficiency. The deep learning-based model outperforms traditional routing protocols, such as AODV and DSR, across several key performance metrics. Peaks measured during medium node density conditions revealed the deep learning model provided 7.1 Mbps of throughput which exceeded AODV's 5.0 Mbps and DSR's 5.3 Mbps capacity. The performance of deep learning model exceeded both AODV (87.1%) and DSR (89.4%) since it delivered packets with 95.2% effectiveness in low node density environments. The deep learning method exhibited lower average end-to-end delay levels of 110 milliseconds in medium node density conditions but displayed average end-to-end delays of 130 milliseconds for AODV and 125 milliseconds for DSR. The deep learning model brought about decreased energy usage which plays a vital role in Worldwide ATM Network developments. In terms of energy consumption, the model reached 2.3 Joules during low node density conditions better than AODV's 3.0 Joules and DSR's 2.8 Joules. The outcomes reveal how deep learning models optimize routing decisions for resource-limited dynamic ad hoc networks through enhancements of transmission speed and packet delivery ratio with decreased latency and reduced power expenditure. Deep learning systems bring intense potential as a solution to fight WANET vulnerabilities driven by dynamic environments and restricted capacities. The CNN-LSTM model demonstrates its ability to boost network performance which positions it as an outstanding solution for applications including disaster recovery and military operations and remote communication that depend on system reliability with low latency and energy efficiency.

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